27) Prove that $P(A) \subseteq P(B)$ if and only if $A \subseteq B$.

11 **Solution:** Let, $P(A) \subseteq P(B)$. By Defn. 3, $\forall x (x \in P(A) \rightarrow x \in P(B))$

a By Defn. 6, we know that $P(A)$ and $P(B)$ are the set of all subsets of A and

b B respectively. By Theorem 1, $A \subseteq A$. $\boxed{\therefore A \in P(A)}$. From Defn. 3,

c $A \in P(B)$. From Defn. 6, we can say that $A \subseteq B$.

d Let, $A \subseteq B$. By Defn. 3, $\forall x (x \in A \rightarrow x \in B)$.

By Defn. 6, we know that $P(A)$ and $P(B)$ are the set of all subsets of A and

e B respectively. By Theorem 1, $B \subseteq B$. $\boxed{\therefore B \in P(B)}$.

f Let, C be an arbitrary member of $P(A)$. By Defn 3, $C \subseteq A$. From Ex-19, $C \subseteq B$. By Defn. 3, $C \in P(B)$.

g $\therefore$ Any arbitrary member of $P(A)$ must be a member of $P(B)$. By Defn. 3, $\boxed{P(A) \subseteq P(B)}$ □

12 28) Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.

a **Solution:** By Defn. 3, $\forall x (x \in A \rightarrow x \in C)$. By Defn. 3, $\forall y (y \in B \rightarrow y \in D)$.

b) $A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$ (By Defn. 8). $C \times D = \{(c,d) \mid c \in C \wedge d \in D\}$ (By Defn. 8)

c)

d Let, (p,q) be an arbitrary element of $A \times B$. $\therefore$ From Defn. 8, $p \in A$ and $q \in B$. From Defn. 3, $p \in C$ and $q \in D$. By Defn. 8, $(p,q) \in C \times D$.

g $\therefore$ Any arbitrary element of $A \times B$ is also an element of $C \times D$. By Defn. 3, $\boxed{A \times B \subseteq C \times D}$ □

f 29) Let $A = \{a,b,c,d\}$ and $B = \{y,z\}$. Find $A \times B$ and $B \times A$

f $A \times B = \{(a,y),(a,z),(b,y),(b,z),(c,y),(c,z),(d,y),(d,z)\}$

$B \times A = \{(y,a),(y,b),(y,c),(y,d),(z,a),(z,b),(z,c),(z,d)\}$

f 34) Let $A = \{a,b,c\}$, $B = \{x,y\}$ and $C = \{0,1\}$. Find

a) $A \times B \times C$ $\{(a,x,0),(a,x,1),(a,y,0),(a,y,1),(b,x,0),(b,x,1),(b,y,0),(b,y,1),(c,x,0),(c,x,1),(c,y,0),(c,y,1)\}$